

Comparing and Visualizing Dendrograms

DANA 4840

Read textbook for more details

Prepare Data

```
df <- scale(USArrests)
```

```
# Subset containing 10 rows
```

```
set.seed(123)
```

```
ss <- sample(1:50, 10)
```

```
df <- df[ss,]
```

```
library(dendextend)
```

Comparing Dendrograms

```
## Warning: package 'dendextend' was built under R version 4.0.5
```

```
##
```

```
## -----
```

```
## Welcome to dendextend version 1.15.1
```

```
## Type citation('dendextend') for how to cite the package.
```

```
##
```

```
## Type browseVignettes(package = 'dendextend') for the package vignette.
```

```
## The github page is: https://github.com/talgalili/dendextend/
```

```
##
```

```
## Suggestions and bug-reports can be submitted at: https://github.com/talgalili/dendextend/issues
```

```
## Or contact: <tal.galili@gmail.com>
```

```
##
```

```
## To suppress this message use: suppressPackageStartupMessages(library(dendextend))
```

```
## -----
```

```
##
```

```
## Attaching package: 'dendextend'
```

```
## The following object is masked from 'package:stats':
```

```
##
```

```
## cutree
```

```
# Compute distance matrix
```

```
res.dist <- dist(df, method = "euclidean")
```

```
# Compute 2 hierarchical clusterings using 2 methods
```

```
hc1 <- hclust(res.dist, method = "average")
```

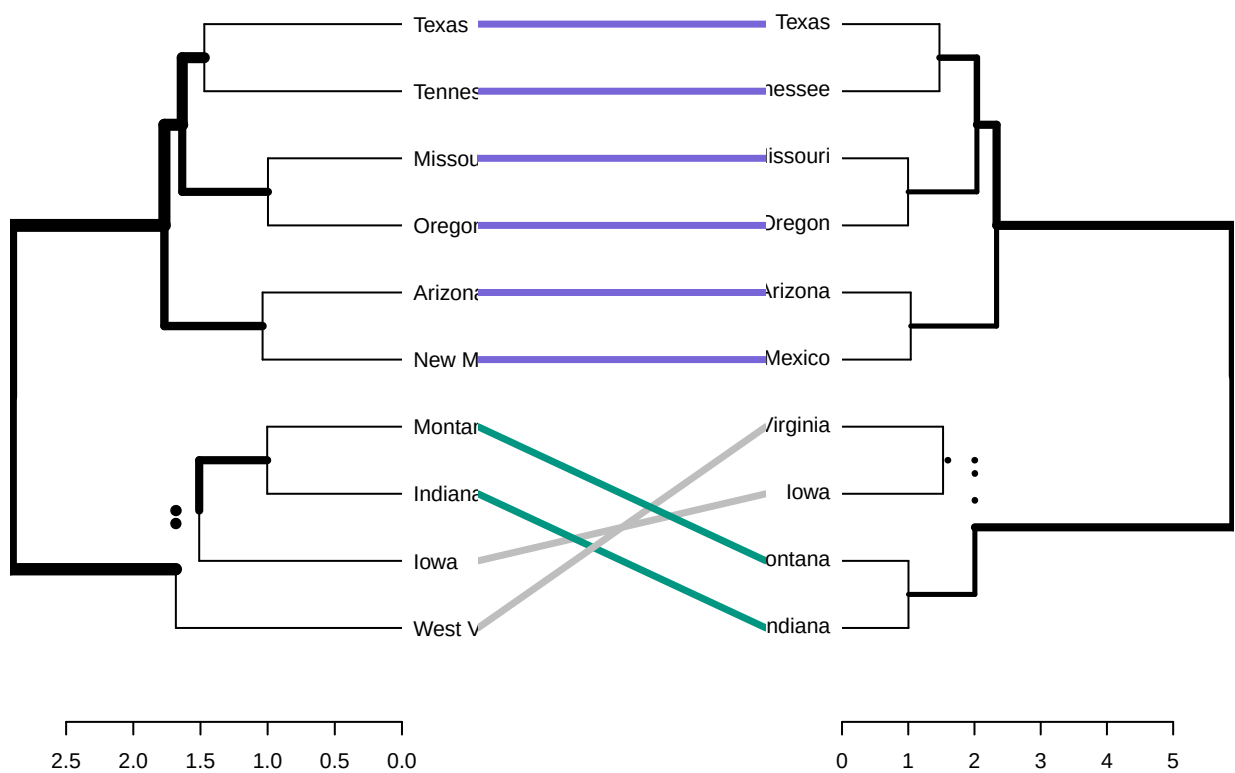
```
hc2 <- hclust(res.dist, method = "ward.D2")
```

```
# Create two dendrograms
dend1 <- as.dendrogram (hc1)
dend2 <- as.dendrogram (hc2)

# Create a list to hold dendrograms
dend_list <- dendlist(dend1, dend2)
```

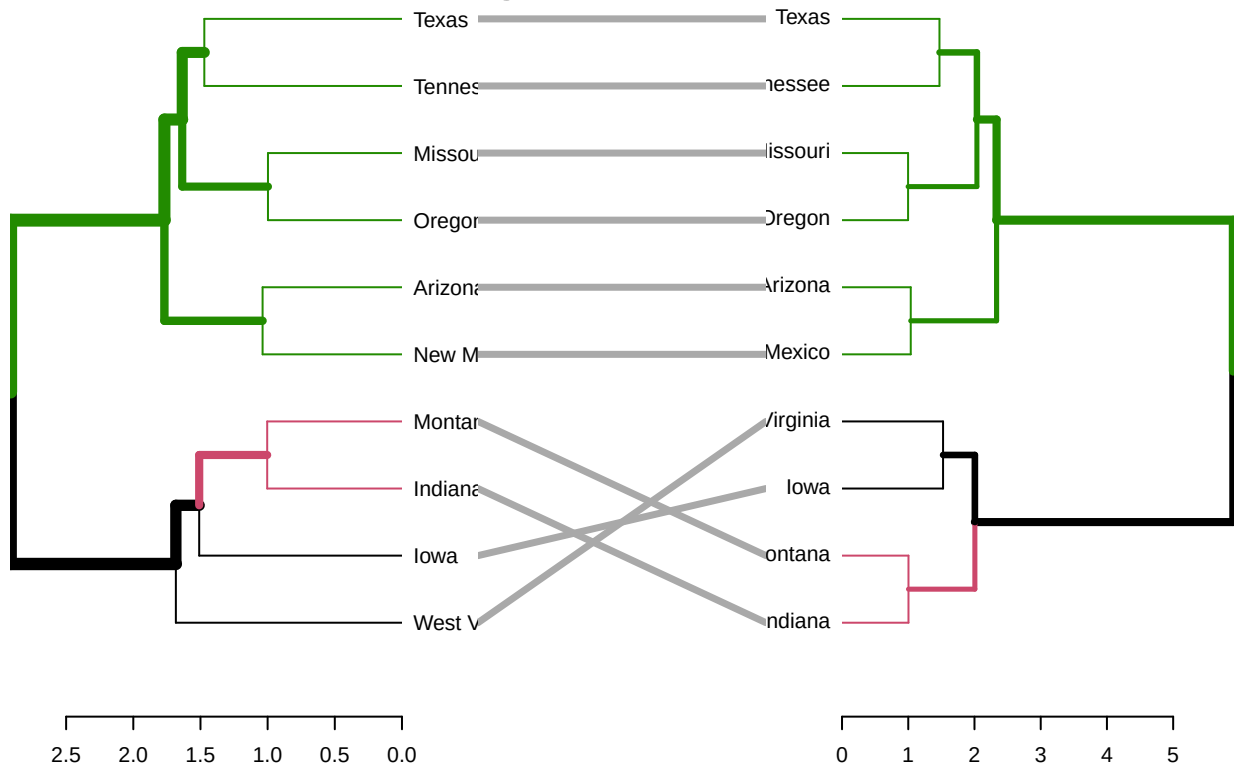
Visual Comparison of two dendrograms

```
tanglegram(dend1, dend2)
```



```
#Customized the tanglegram using many other options as follow:
tanglegram(dend1, dend2,
highlight_distinct_edges = FALSE, # Turn-off dashed lines
common_subtrees_color_lines = FALSE, # Turn-off line colors
common_subtrees_color_branches = TRUE, # Color common branches
main = paste("entanglement =", round(entanglement(dend_list), 2))
) # Alignment quality
```

entanglement = 0.09



Note that, “unique” nodes, with a combination of labels/items not present in the other tree, are highlighted with dashed lines.

Correlation matrix between a list of dendrograms The function `cor.dendlist()` is used to compute “Baker” or “Cophenetic” correlation matrix between a list of trees. The value can range between -1 to 1. With near 0 values meaning that the two trees are not statistically similar.

```
# Cophenetic correlation matrix
cor.dendlist(dend_list, method = "cophenetic")
```

```
##           [,1]      [,2]
## [1,] 1.0000000 0.9925544
## [2,] 0.9925544 1.0000000
```

```
# Baker correlation matrix
cor.dendlist(dend_list, method = "baker")
```

```
##           [,1]      [,2]
## [1,] 1.0000000 0.9895528
## [2,] 0.9895528 1.0000000
```

```
# Cophenetic correlation coefficient
cor_cophenetic(dend1, dend2)
```

```
## [1] 0.9925544
```

```
# Baker correlation coefficient
cor_bakers_gamma(dend1, dend2)
```

```
## [1] 0.9895528
```

It's also possible to compare simultaneously multiple dendrograms. A chaining operator `%>%` is used to run multiple function at the same time. It's useful for simplifying the code:

```
# Create multiple dendrograms by chaining
dend1 <- df %>% dist %>% hclust("complete") %>% as.dendrogram
dend2 <- df %>% dist %>% hclust("single") %>% as.dendrogram
dend3 <- df %>% dist %>% hclust("average") %>% as.dendrogram
dend4 <- df %>% dist %>% hclust("centroid") %>% as.dendrogram
```

```
# Compute correlation matrix
dend_list <- dendlist("Complete" = dend1, "Single" = dend2,
                     "Average" = dend3, "Centroid" = dend4)
```

```
cors <- cor.dendlist(dend_list)
# Print correlation matrix
round(cors, 2)
```

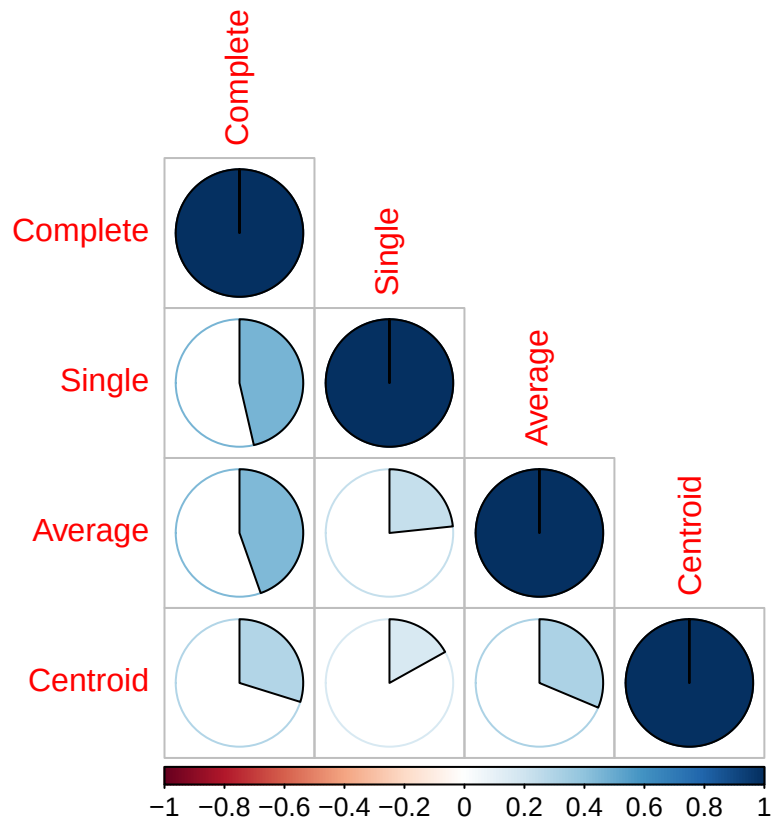
```
##           Complete Single Average Centroid
## Complete      1.00    0.46    0.45    0.30
## Single        0.46    1.00    0.23    0.17
## Average       0.45    0.23    1.00    0.31
## Centroid      0.30    0.17    0.31    1.00
```

```
# Visualize the correlation matrix using corrplot package
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.0.4
```

```
## corrplot 0.84 loaded
```

```
corrplot(cors, "pie", "lower")
```



Visualizing Dendrograms

```
# Load data
data(USArrests)

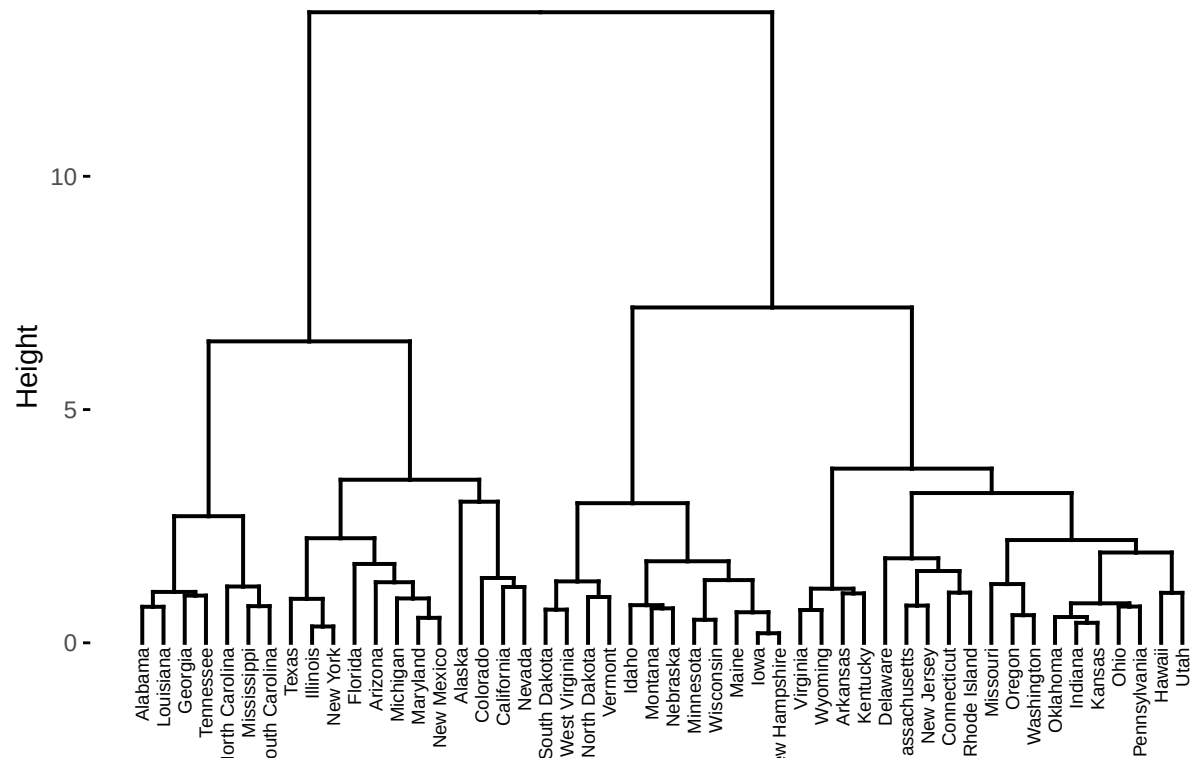
# Compute distances and hierarchical clustering
dd <- dist(scale(USArrests), method = "euclidean")
hc <- hclust(dd, method = "ward.D2")

## install.packages(c("factoextra", "dendextend"))
library(factoextra)

## Warning: package 'factoextra' was built under R version 4.0.5
## Loading required package: ggplot2
## Warning: package 'ggplot2' was built under R version 4.0.5
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
fviz_dend(hc, cex = 0.5)
```

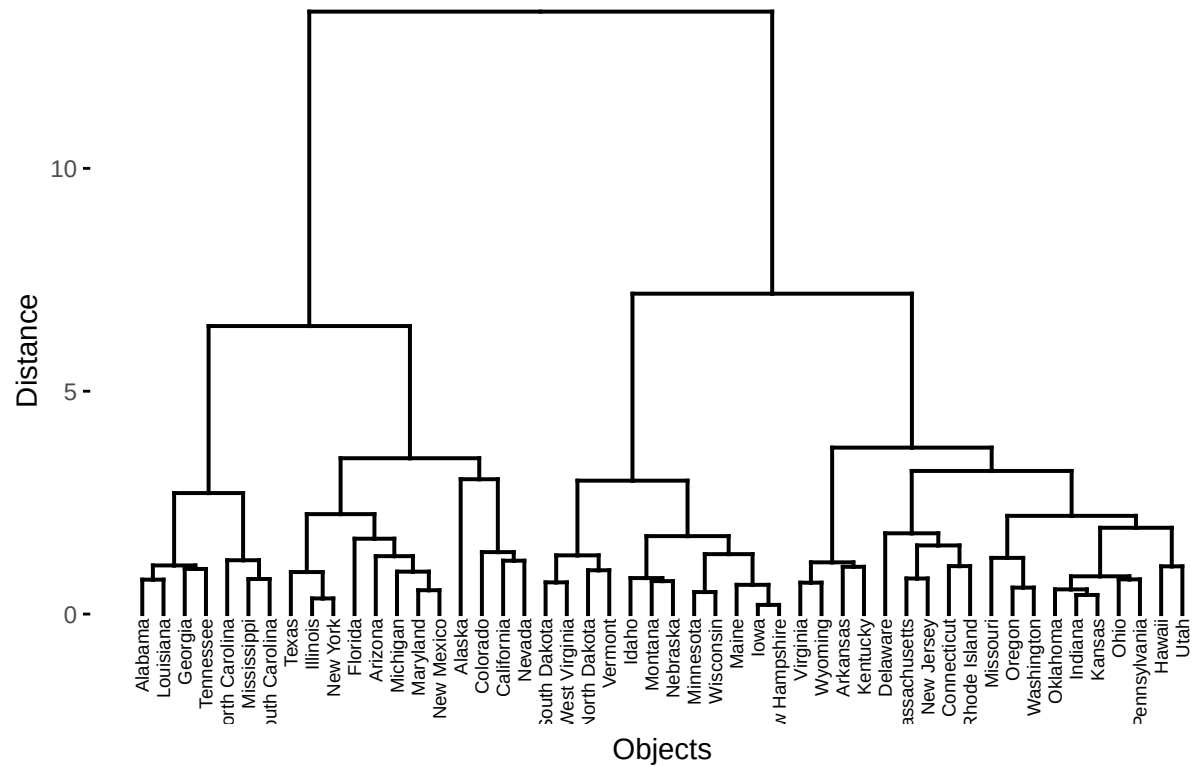
Cluster Dendrogram



You can use the arguments main, sub, xlab, ylab to change plot titles as follow:

```
fviz_dend(hc, cex = 0.5,
           main = "Dendrogram - ward.D2",
           xlab = "Objects", ylab = "Distance", sub = "")
```

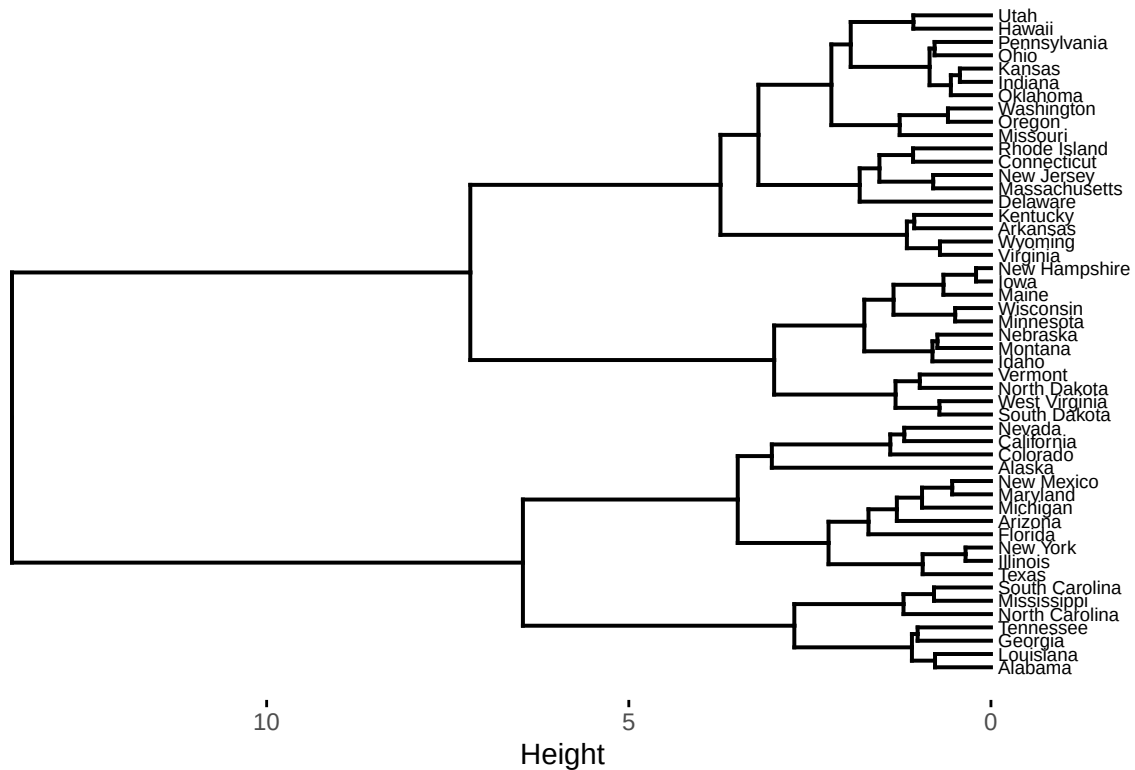
Dendrogram – ward.D2



To draw a horizontal dendrogram, type this:

```
fviz_dend(hc, cex = 0.5, horiz = TRUE)
```

Cluster Dendrogram



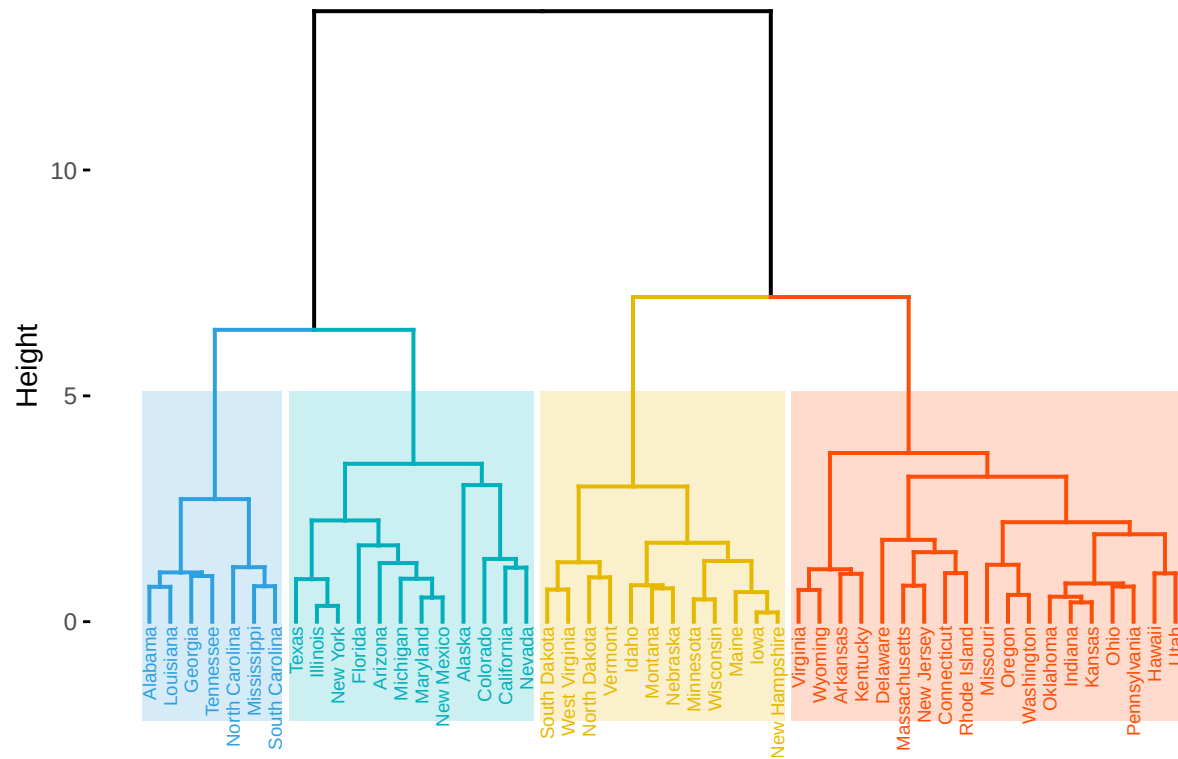
It's also possible to cut the tree at a given height for partitioning the data into multiple groups as described in the previous chapter: Hierarchical clustering. In this case, it's possible to color branches by groups and to add rectangle around each group.

For example:

```
fviz_dend(hc, k = 4, # Cut in four groups
  cex = 0.5, # label size
  k_colors = c("#2E9FDF", "#00AFBB", "#E7B800", "#FC4E07"),
  color_labels_by_k = TRUE, # color labels by groups
  rect = TRUE, # Add rectangle around groups
  rect_border = c("#2E9FDF", "#00AFBB", "#E7B800", "#FC4E07"),
  rect_fill = TRUE)
```

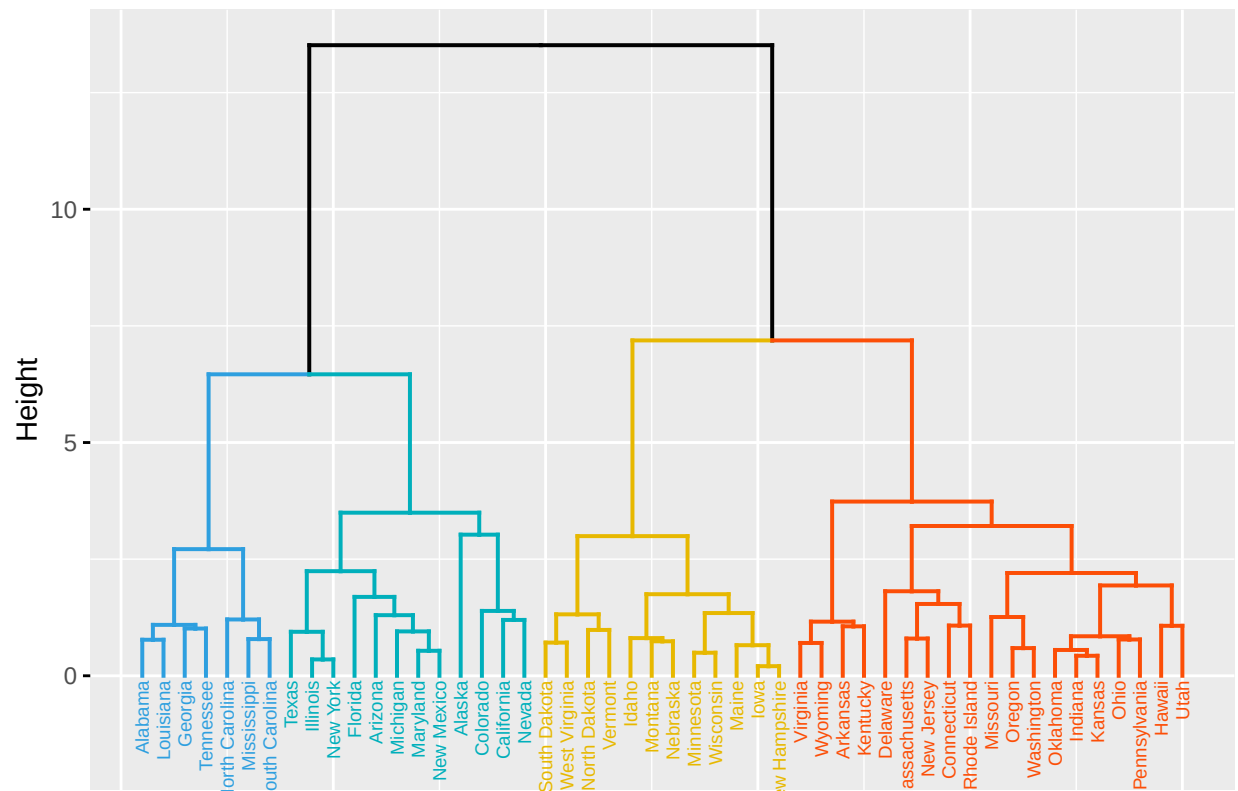
```
## Warning in if (color == "cluster") color <- "default": the condition has length
## > 1 and only the first element will be used
```


Cluster Dendrogram



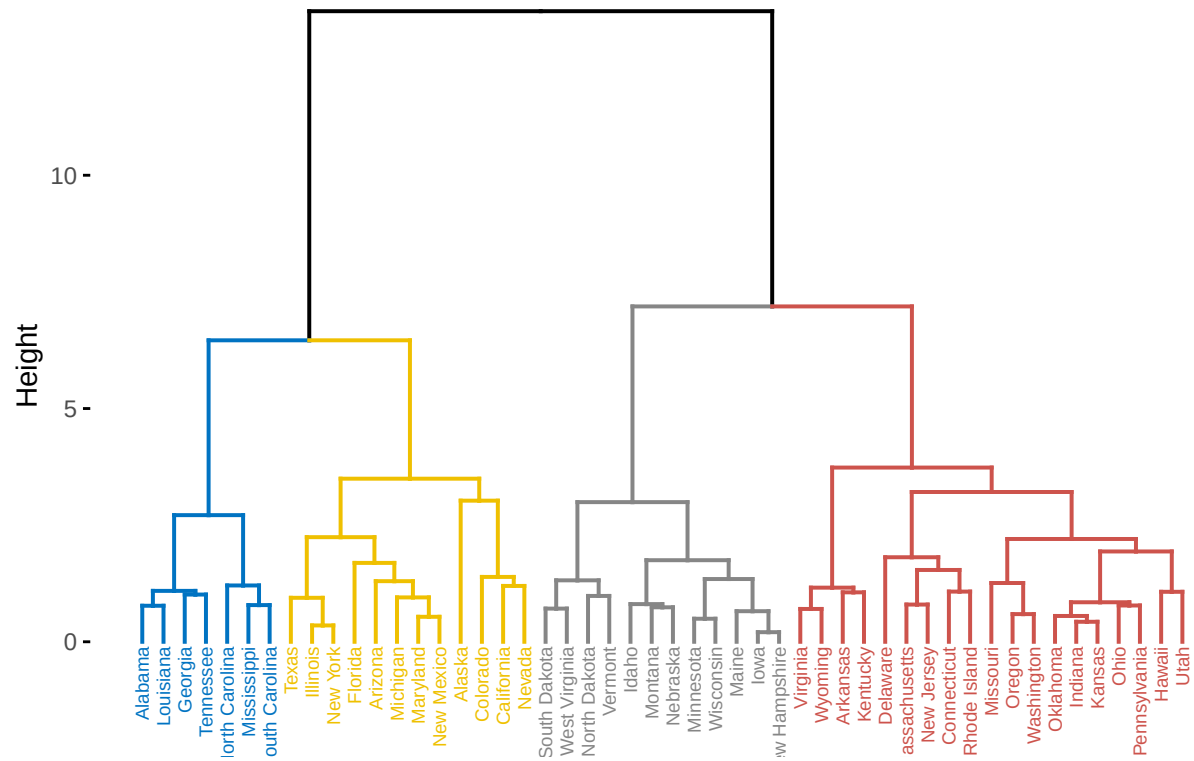
```
fviz_dend(hc, k = 4,                                # Cut in four groups
  cex = 0.5,                                          # label size
  k_colors = c("#2E9FDF", "#00AFBB", "#E7B800", "#FC4E07"),
  color_labels_by_k = TRUE, # color labels by groups
  ggtheme = theme_gray() # Change theme
)
```

Cluster Dendrogram



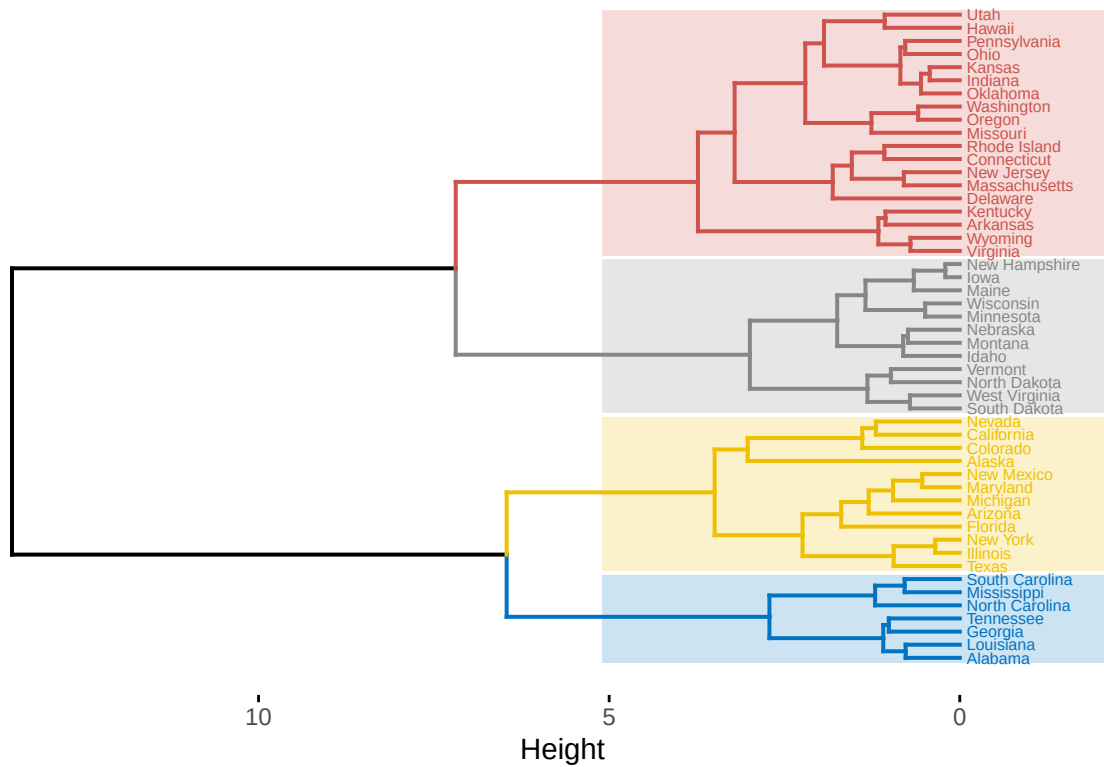
```
fviz_dend(hc, cex = 0.5, k = 4, # Cut in four groups
          k_colors = "jco")
```

Cluster Dendrogram

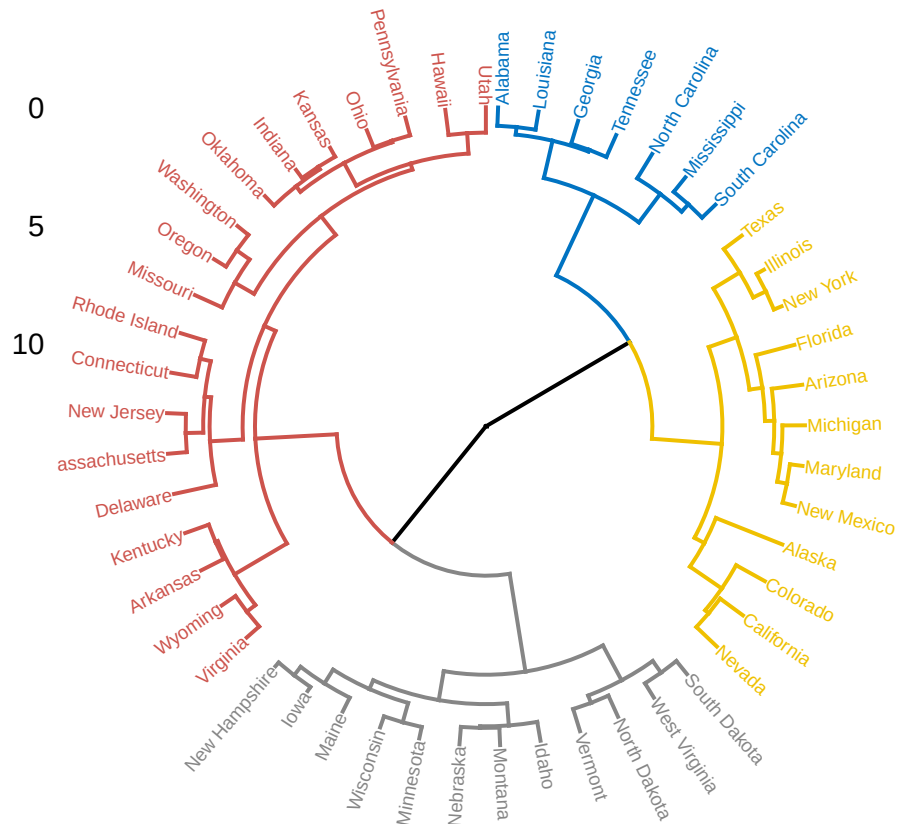


```
fviz_dend(hc, k = 4, cex = 0.4, horiz = TRUE, k_colors = "jco",
  rect = TRUE, rect_border = "jco", rect_fill = TRUE)
```

Cluster Dendrogram



```
fviz_dend(hc, cex = 0.5, k = 4,  
          k_colors = "jco", type = "circular")
```



To plot a phylogenetic-like tree, use `type = "phylogenetic"` and `repel = TRUE` (to avoid labels overplotting). This functionality requires the R package *igraph*.

```
require("igraph")
```

```
## Loading required package: igraph
```

```
##
```

```
## Attaching package: 'igraph'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

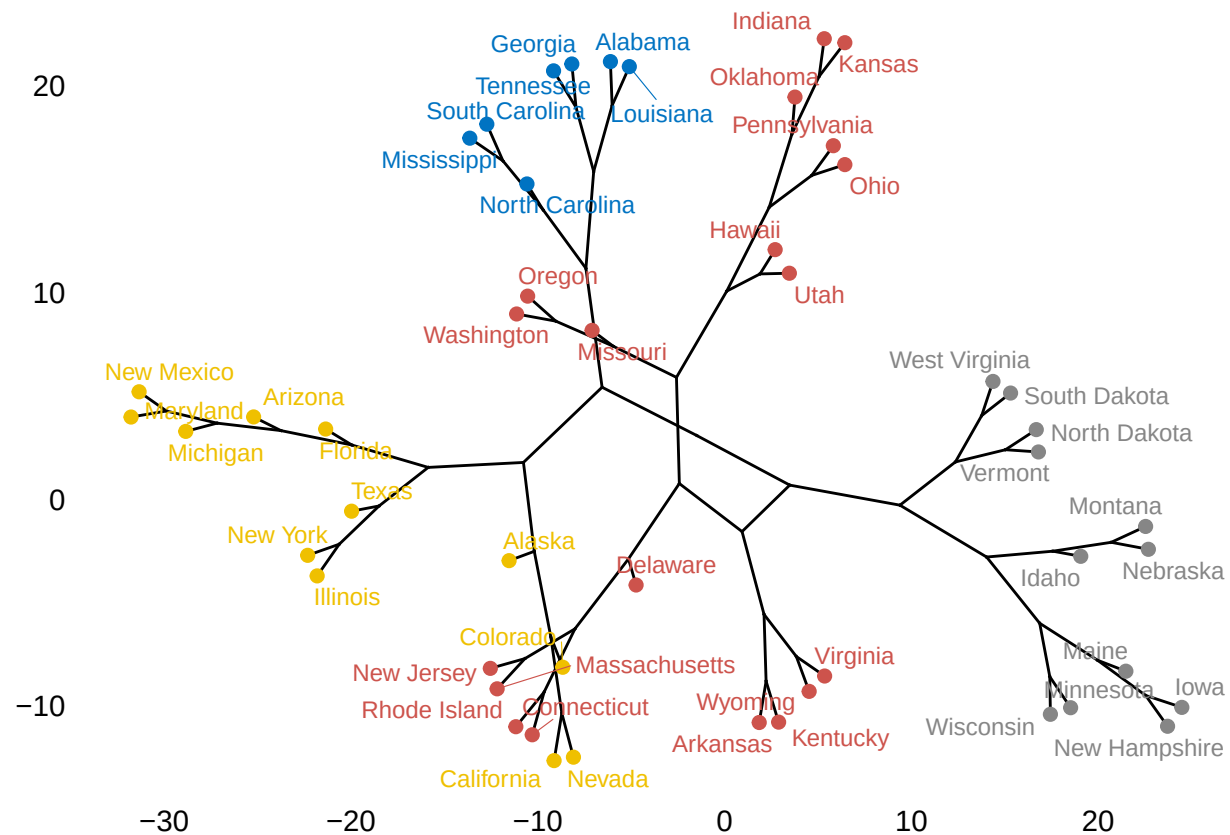
```
## decompose, spectrum
```

```
## The following object is masked from 'package:base':
```

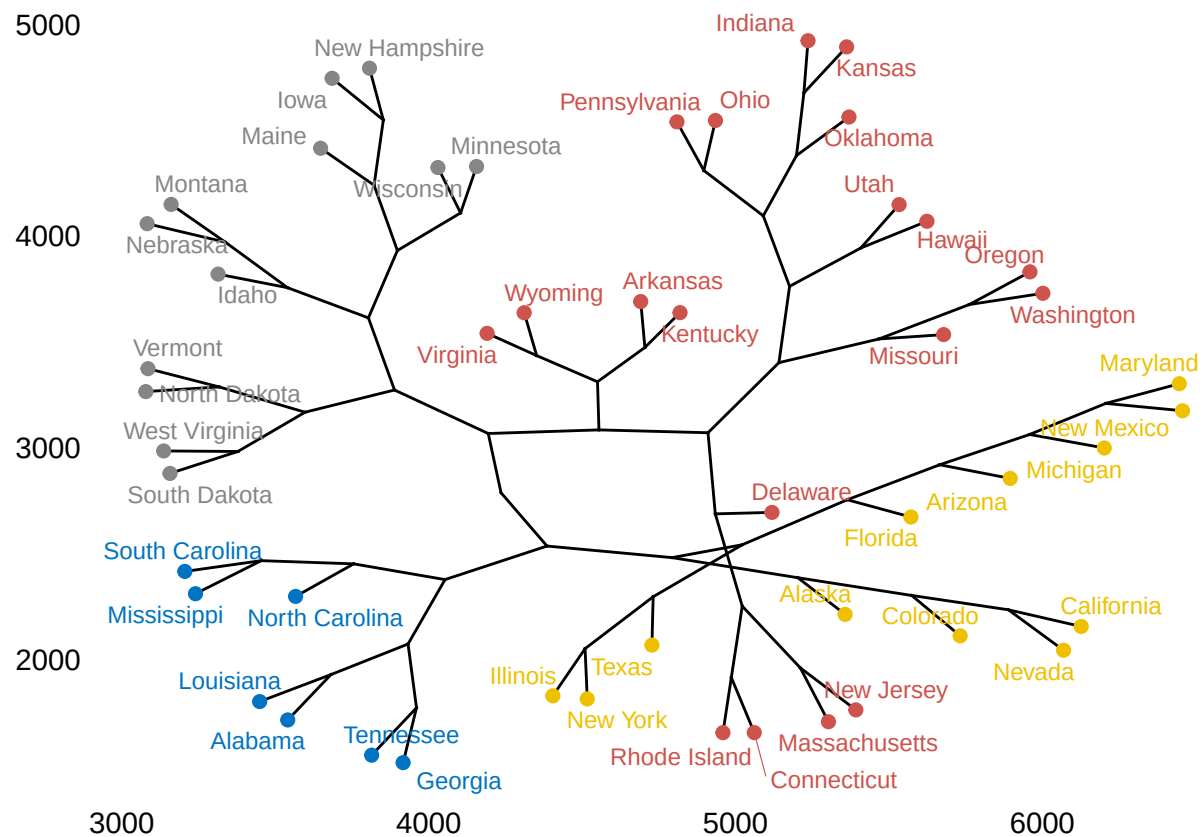
```
##
```

```
## union
```

```
fviz_dend(hc, k = 4, k_colors = "jco",
          type = "phylogenetic", repel = TRUE)
```

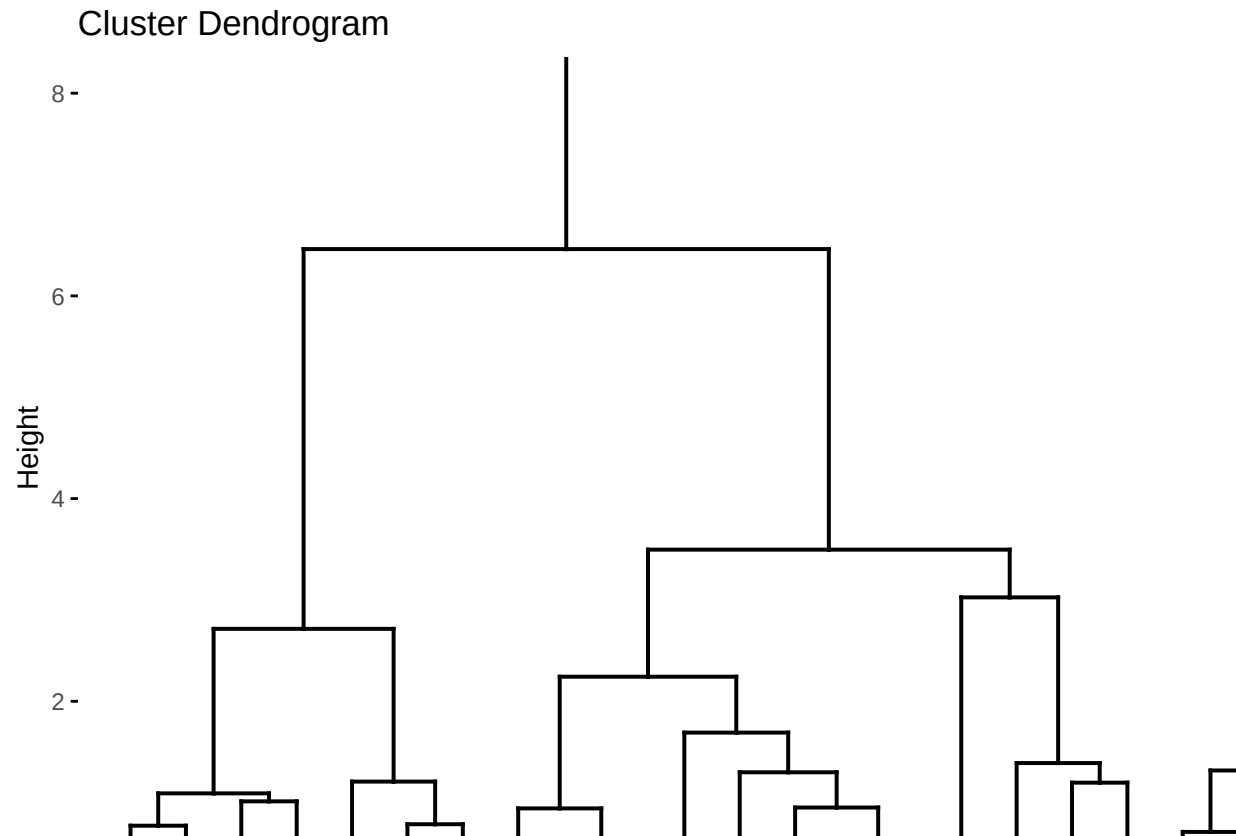


```
require("igraph")
fviz_dend(hc, k = 4, # Cut in four groups
  k_colors = "jco",
  type = "phylogenetic", repel = TRUE,
  phylo_layout = "layout.gem")
```



Zooming in the dendrogram

```
fviz_dend(hc, xlim = c(1, 20), ylim = c(1, 8))
```



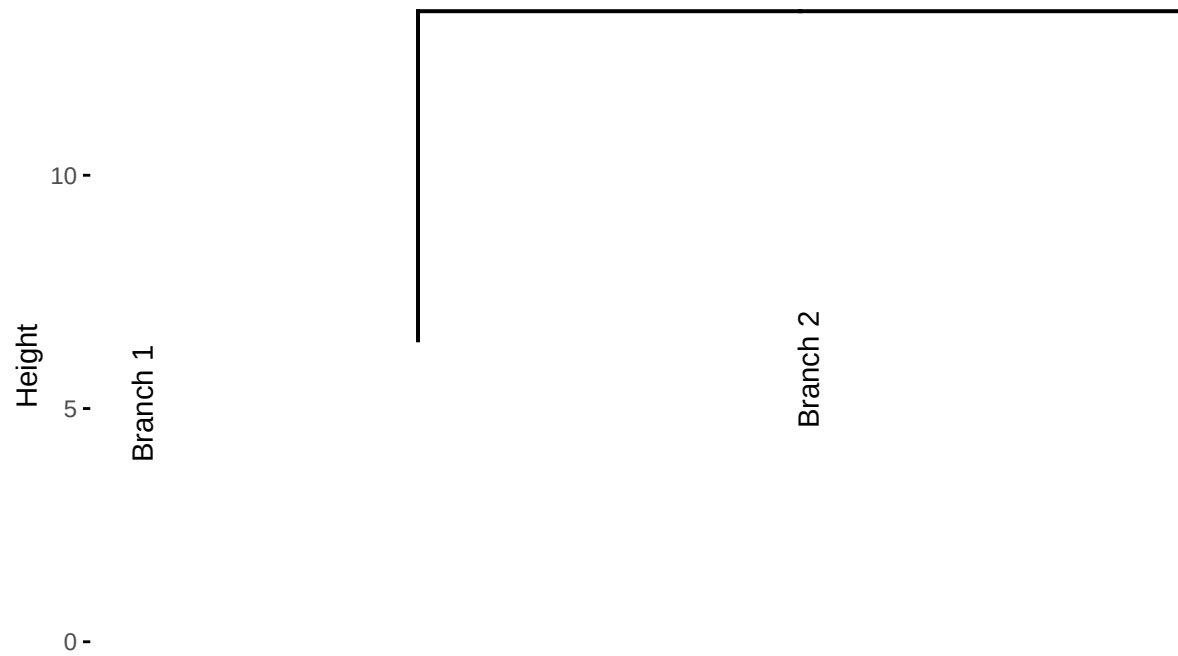
Plotting a sub-tree of dendrograms

```
# Create a plot of the whole dendrogram,
# and extract the dendrogram data
dend_plot <- fviz_dend(hc, k = 4, # Cut in four groups
  cex = 0.5, # label size
  k_colors = "jco"
)
dend_data <- attr(dend_plot, "dendrogram") # Extract dendrogram data
```

```
# Cut the dendrogram at height h = 10
dend_cuts <- cut(dend_data, h = 10)
# Visualize the truncated version containing
# two branches
fviz_dend(dend_cuts$upper)
```

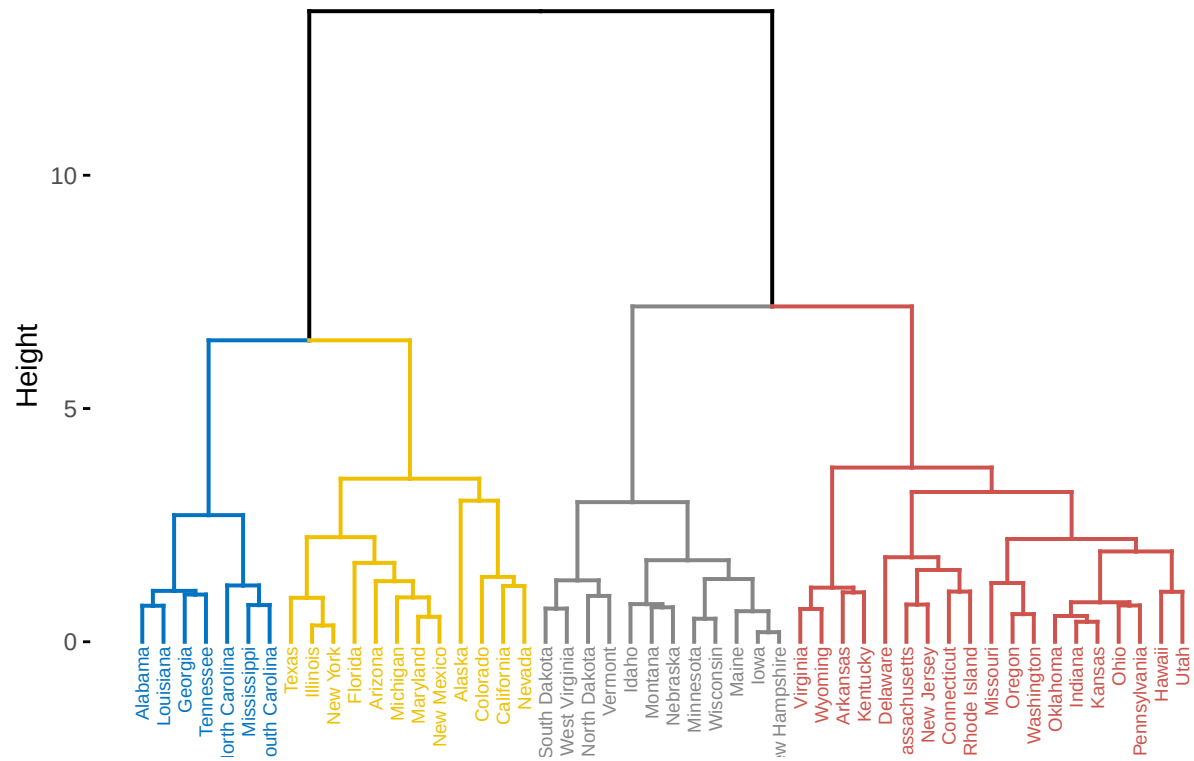
```
## Warning in min(-diff(our_dend_heights)): no non-missing arguments to min;
## returning Inf
```


Cluster Dendrogram



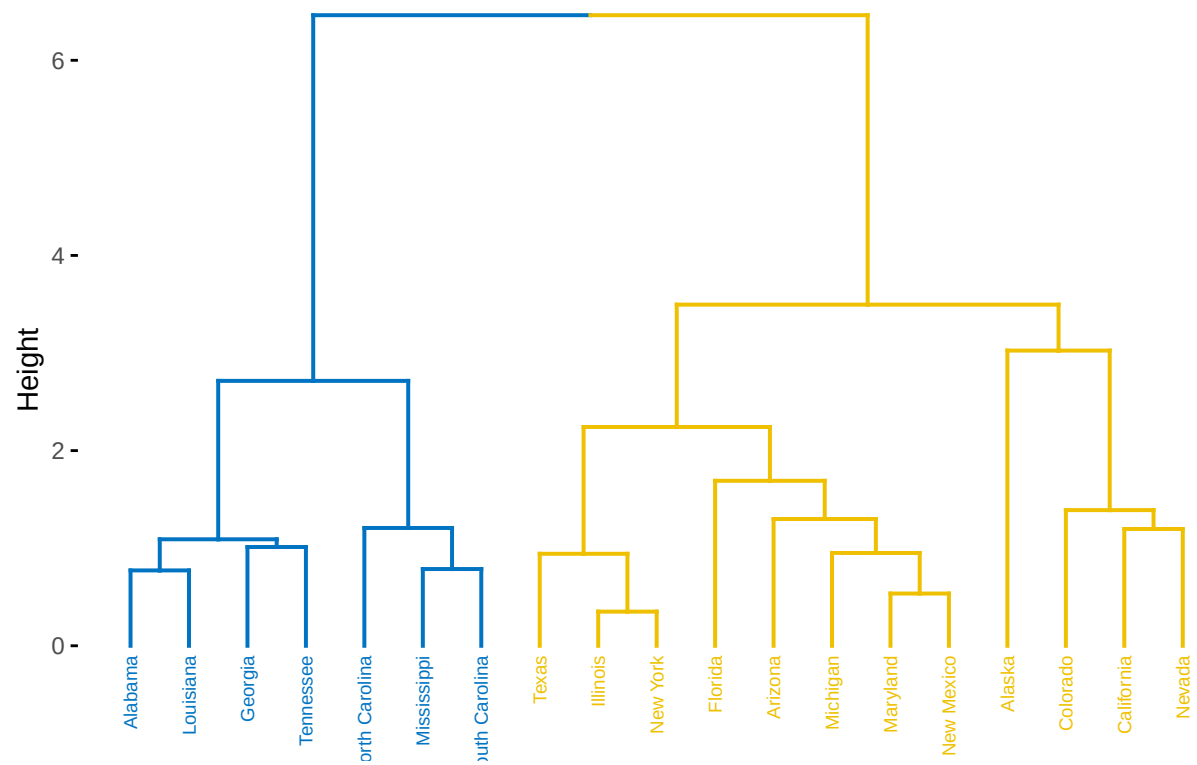
```
# Plot the whole dendrogram  
print(dend_plot)
```

Cluster Dendrogram



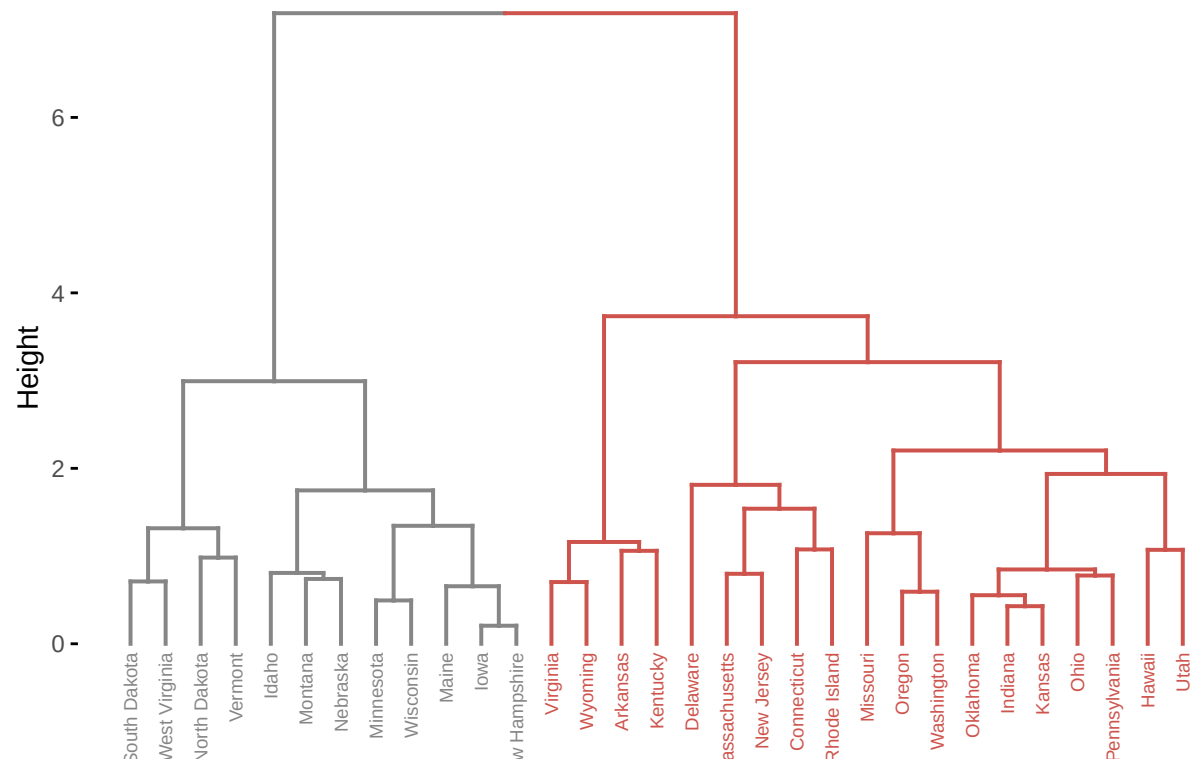
```
# Plot subtree 1
fviz_dend(dend_cuts$lower[[1]], main = "Subtree 1")
```

Subtree 1

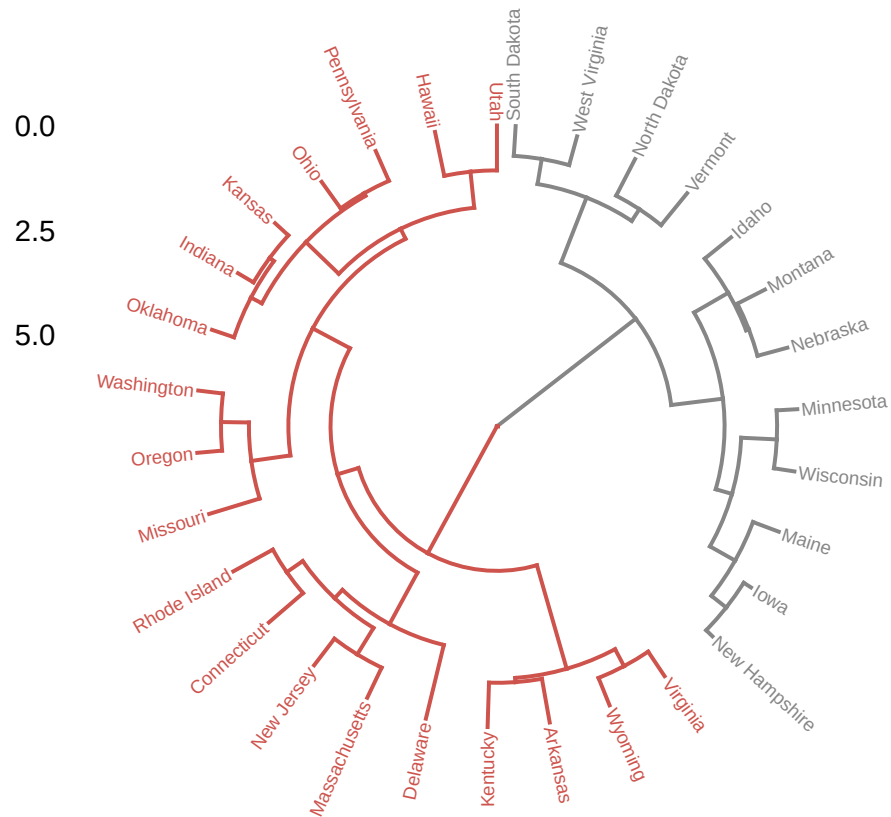


```
# Plot subtree 2
fviz_dend(dend_cuts$lower[[2]], main = "Subtree 2")
```

Subtree 2



```
fviz_dend(dend_cuts$lower[[2]], type = "circular")
```



Saving dendrogram into a large PDF page

```
## pdf("dendrogram.pdf", width=30, height=15)           # Open a PDF
## p <- fviz_dend(hc, k = 4, cex = 1, k_colors = "jco" ) # Do plotting
## print(p)
## dev.off()                                           # Close the PDF
```